```
import java.util.Arrays;
import java.util.Random;

public class SelectionSort {

	// Sorting method
	public static void sort(int[] arr) {

		for (int i = 0; i < arr.length - 1; i++) {
			int minIndex = i;

			for (int j = i + 1; j < arr.length; j++) {
				if (arr[j] < arr[minIndex]) {
					minIndex = j;
				}
			}

			int temp = arr[i];
			arr[i] = arr[minIndex];
			arr[minIndex] = temp;
		}
	}

	// Run this algorithm individually
	public static void main(String[] args) {

		int[] arr = new int[1000];
		Random random = new Random();

		for (int i = 0; i < arr.length; i++) {
			arr[i] = random.nextInt(10000);
		}

		System.out.println("Before Sorting:");
		System.out.println(Arrays.toString(arr));

		long startTime = System.nanoTime();
		sort(arr);
		long endTime = System.nanoTime();

		System.out.println("After Selection Sort:");
		System.out.println(Arrays.toString(arr));

		System.out.println("Execution Time: "
				+ (endTime - startTime) + " nanoseconds");
	}
}
```

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